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MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER OCTOBER/NOVEMBER 2025 (TERM ID 2530)

BAC6434 – ADVANCED MANAGEMENT ACCOUNTING

(All sections / Groups)

03 MARCH 2026
9.00 a.m. to 12.00 p.m.
(3 Hours)

INSTRUCTIONS TO STUDENT

1. This question paper consists of 7 pages excluding the cover page with 4 questions only.
2. Attempt ALL questions. All questions carry equal marks and the distribution of the marks for each question is given.
3. Please write all your answers in the Answer Booklet provided.

QUESTION 1**Part A**

Tech Systems Sdn Bhd, (Tech) a dynamic and fast-growing smart home device manufacturer based in Johor Bahru, Malaysia, has been steadily gaining traction in the Southeast Asian market. Known for its innovation and quality, the company has been operating at 70% of its production capacity, leaving room for potential expansion.

One morning, the management team received an unexpected proposal from EcoSmart Inc. (EcoSmart), a Canadian firm specializing in sustainable home automation solutions. EcoSmart, facing a sudden surge in demand due to a government-backed green initiative, urgently needed a reliable partner to supply 80,000 smart thermostats over the next four months. They offered RM22.00 per unit, a price lower than Tech's standard selling price but with the promise of a large volume and potential future collaboration.

Tech's standard cost structure for the smart thermostat is as follows:

- Direct materials: RM6.00
- Direct labor: RM8.00 (0.5 hours per unit)
- Manufacturing overhead: RM10.00 (50% variable)

To fulfill the order, Tech would need to hire temporary supervisors, incurring additional fixed overhead costs of RM12,000 per month. The regular selling price of the thermostat is RM30.00, though recent market pressures have led to a price reduction to RM27.00. Regular sales also include sales commissions of 5% and freight costs of RM1.00 per unit, both of which would not apply to the EcoSmart order, as the client would handle logistics and placed the order directly with Tech's executive team.

The production department confirmed that the order could be accommodated without disrupting current operations. However, the management team must now evaluate whether accepting this special order aligns with the company's financial goals and international strategy.

Required:

- (a) Compute the additional direct labor hours each month to fulfill the EcoSmart order.
(1.5 marks)
- (b) Assess the impact on operating income if Tech accepts the EcoSmart order at RM22.00 per unit. Show calculations.
(5 marks)

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- (c) Determine the minimum unit price Tech could accept without reducing operating income. (1 mark)
- (d) If accepting the order causes a loss of 3,000 regular sales units per month at RM27.00, ascertain the revised breakeven price. (5 marks)
- (e) Identify **four (4)** strategic factors to consider before accepting the order. (4 marks)

Part B

A company, EcoClean Services, provides monthly cleaning services to corporate clients. The company wants to estimate its fixed and variable costs to better manage its budget. The following data was collected over the past six months:

Month	Number of Jobs	Total Cost (RM)
January	120	24,000
February	150	27,000
March	100	22,000
April	180	30,000
May	130	25,000
June	160	28,000

Required:

- (a) Using the High-Low Method, estimate the variable cost per job and the fixed cost for EcoClean Services. (7 marks)
- (b) Calculate the expected total cost if the company perform 140 jobs next month. (1.5 marks)

(Total: 25 marks)

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QUESTION 2

Cermai Health Centre runs two programs: drug addict rehabilitation and aftercare (counseling and support of patients after release from a mental hospital). The centre's budget for 2026 is as follows:

	RM	RM
Professional salaries		
* 4 physicians x RM150,000	600,000	
* 12 psychologists x RM75,000	900,000	
* 16 nurses x RM30,000	<u>480,000</u>	1,980,000
Medical supplies		242,000
Rent and clinic maintenance		138,600
Administrative costs to manage patient charts, food, laundry		484,000
Laboratory services		<u>92,400</u>
Total		<u>2,937,000</u>

Mr. Khairul, the director of the centre, is keen on determining the cost of each program. Khairul compiles the following data, describing employee allocations to individual programs:

	Drug	Aftercare	Total employees
Physicians	4	-	4
Psychologists	4	8	12
Nurses	6	10	16

Khairul has recently become aware of activity-based costing (ABC) as a method to refine costing systems. He asks his accountant, Ms. Gowri, how he should apply this technique. Gowri obtains the following budgeted information for 2026:

	Drug	Aftercare	Total
Square feet of space occupied by each program	9,000	12,000	21,000
Patient-years of service	50	60	110
Number of laboratory tests	1,400	700	2,100

Required:

- (a) Using cost-allocation bases that you believe are the most appropriate for allocating indirect costs to programs, compute the budgeted indirect cost rates for:

(i) medical supplies (2 marks)

(ii) rent and clinic maintenance (2 marks)

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- (iii) administrative costs for patient charts, food, and laundry (2 marks)
- (iv) laboratory services (2 marks)
- (b) Using an activity-based costing approach to cost analysis, compute the budgeted cost of each program and the budgeted cost per patient-year of the drug program. (9 marks)
- (c) Asses the benefits that Cermai Health Centre may obtain by implementing the ABC system. (4 marks)
- (d) Discover factors, other than cost, which Cermai Health Centre should consider in allocating resources to its programs. (2 marks)
- (e) En. Khairul would like to develop a cost control system in Cermai Health Centre, assess how does the cost control system interact with the overall cost management system for this centre. (2 marks)

(Total: 25 marks)

QUESTION 3

Part A

SmartPack Sdn. Bhd. (SmartPac1), a leading Malaysian company in the eco-friendly packaging industry, has built a strong reputation for innovation and sustainability. In response to increasing global demand for environmentally responsible packaging and rising operational costs, the company is evaluating a major capital investment in a new Automated Packaging System (APS). This system is designed to streamline production, reduce material waste, and enhance customer satisfaction through faster delivery times.

The APS is priced at RM150,000,000 and has a projected useful life of 8 years. It is expected to have a salvage value of RM5,000,000, although the company anticipates selling it for RM4,000,000 at the end of its life. SmartPack uses straight-line depreciation for both accounting and tax purposes.

To implement the APS, SmartPack will incur a one-time implementation cost of RM50,000,000, which includes employee training, temporary production downtime, and system integration. This cost will be capitalised along with the APS purchase.

Once operational, the APS is expected to generate the following annual benefits and costs:

- Material waste reduction:

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- RM20,000,000 in savings due to more precise packaging.
- Labour cost savings:
RM12,000,000 from reduced manual labour requirements.
- Additional revenue:
RM10,000,000 from improved delivery speed and customer satisfaction.
- Annual maintenance cost:
RM10,000,000 to ensure optimal performance.

Additionally, the project requires an increase in working capital of RM6,000,000, which will be fully recovered at the end of year 8. SmartPack's required rate of return is 12%, and the corporate tax rate is 25%. All cash flows are assumed to occur at the end of each year, except for the initial investment, which occurs at the start of the project.

Required:

- (a) Apply your knowledge of capital budgeting to determine the relevant after-tax cash flows for the APS investment at each of the following stages:
- (i) Project initiation (2 marks)
 - (ii) Project operations (6 marks)
 - (iii) Project disposal (3 marks)
- (b) Ascertain the Net Present Value (NPV) of the APS investment and recommend whether SmartPack should proceed with the investment. (5 marks)
- (c) Calculate the after-tax payback period for the APS. (1 mark)
- (d) Present **two (2)** limitations of using discounted cash flow (DCF) analysis in this context. (1 mark)
- (e) Illustrate **two (2)** additional factors SmartPack should consider before making the investment decision. (1 mark)

Part B

You are the budget coordinator for a mid-sized manufacturing company. The Chief Executive Officer (CEO) has asked you to redesign the budgeting process to improve

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employee engagement and budget accuracy. You plan to implement a participative budgeting approach, allowing department heads to contribute to their own budget plans.

Required:

Present to the CEO **two (2)** significance of participative budgeting and one challenge in its implementation.

(6 marks)

(Total: 25 marks)

QUESTION 4

Part A

Gambus Bhd. operates two investment divisions in Malaysia, one in Selangor and another in Perak.

In each of the following scenarios, the Selangor Division may sell all of its products externally, or some internally to the Perak Division and the remaining to outside clients. The Selangor Division has a production capacity of 400,000 units annually. The data for each independent case are as follows:

Selangor Division	Case 1 (RM)	Case 2 (RM)
Selling price to outside customers	125	110
Production costs per unit:		
Direct material	40	30
Direct labour	20	18
Variable overhead	13	12
Fixed overhead (based on capacity)	11	11
Other variable selling and delivery costs per unit*	16	13

*In either case, RM11 of the selling expenses will not be incurred on intracompany transfers.

Perak Division	Case 1	Case 2
Number of speakers needed annually	80,000	80,000
Current unit price being paid to outside supplier	RM110	RM112

Required:

- (a) For each case, compute the upper and lower limits for a transfer price for the products. Interpret your answers.

(10 marks)

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- (b) For each case, compute a transfer price for the Selangor Division that will provide a RM10 contribution margin per unit.
(2 marks)
- (c) Interpret transfer pricing and assess its purpose for companies. In what ways might the application of transfer pricing enhance or hinder goal congruence?
(4 marks)
- (d) What is the primary basis for evaluating the performance of the manager of an investment centre? Show the formula for this basis.
(2 marks)
- (e) In your opinion, assess two behavioural principles that pertain to (i) the manager being evaluated and (ii) top management.
(2 marks)

Part B

Illustrate the role of benchmarking in evaluating managers.

(5 marks)

(Total: 25 marks)

End of Paper

APPENDIX

Present Value of \$1.00

$$P = \frac{S}{(1 + r)^n} \text{ In this table } S = \$1.00.$$

Periods	2%	4%	6%	8%	10%	12%	14%	16%	18%
1	0.980	0.962	0.943	0.926	0.909	0.893	0.877	0.862	0.847
2	0.961	0.925	0.890	0.857	0.826	0.797	0.769	0.743	0.718
3	0.942	0.889	0.840	0.794	0.751	0.712	0.675	0.641	0.609
4	0.924	0.855	0.792	0.735	0.683	0.636	0.592	0.552	0.516
5	0.906	0.822	0.747	0.681	0.621	0.567	0.519	0.476	0.437
6	0.888	0.790	0.705	0.630	0.564	0.507	0.456	0.410	0.370
7	0.871	0.760	0.665	0.583	0.513	0.452	0.400	0.354	0.314
8	0.853	0.731	0.627	0.540	0.467	0.404	0.351	0.305	0.266
9	0.837	0.703	0.592	0.500	0.424	0.361	0.308	0.263	0.225
10	0.820	0.676	0.558	0.463	0.386	0.322	0.270	0.227	0.191

Present Value of Annuity \$1.00 in Arrears*

$$P_n = \frac{1}{r} \left[1 - \frac{1}{(1 + r)^n} \right]$$

Periods	2%	4%	6%	8%	10%	12%	14%	16%	18%
1	0.980	0.962	0.943	0.926	0.909	0.893	0.877	0.862	0.847
2	1.942	1.886	1.833	1.783	1.736	1.690	1.647	1.605	1.566
3	2.884	2.775	2.673	2.577	2.487	2.402	2.322	2.246	2.174
4	3.808	3.630	3.465	3.312	3.170	3.037	2.914	2.798	2.690
5	4.713	4.452	4.212	3.993	3.791	3.605	3.433	3.274	3.127
6	5.601	5.242	4.917	4.623	4.355	4.111	3.889	3.685	3.498
7	6.472	6.002	5.582	5.206	4.868	4.564	4.288	4.039	3.812
8	7.325	6.733	6.210	5.747	5.335	4.968	4.639	4.344	4.078
9	8.162	7.435	6.802	6.247	5.759	5.328	4.946	4.607	4.303
10	8.983	8.111	7.360	6.710	6.145	5.650	5.216	4.833	4.494

